

Assignment 6

Q1. Write a menu driven program with 4 options (Insert, Delete, Display, and Exit) to demonstrate the working of Queues using arrays.

```
#include <iostream>
using namespace std;

#define SIZE 5

int queue[SIZE];
int front = -1;
int rear = -1;

void insert()
{
    int x;

    if (rear == SIZE - 1)
    {
        cout << "Queue is full\n";
        return;
    }

    cout << "Enter value: ";
    cin >> x;

    if (front == -1)
    {
        front = 0;
    }

    rear++;
    queue[rear] = x;

    cout << "Element inserted\n";
}

void deleteElement()
{
    if (front == -1 || front > rear)
    {
        cout << "Queue is empty\n";
    }
}
```

```

        return;
    }

    cout << "Deleted element: " << queue[front] << endl;

    front++;

    if (front > rear)
    {
        front = -1;
        rear = -1;
    }
}

void display()
{
    if (front == -1)
    {
        cout << "Queue is empty\n";
        return;
    }

    cout << "Queue: ";

    for (int i = front; i <= rear; i++)
    {
        cout << queue[i] << " ";
    }

    cout << endl;
}

int main()
{
    int choice;

    do
    {
        cout << "\n QUEUE MENU \n";
        cout << "1. Insert\n";
        cout << "2. Delete\n";
        cout << "3. Display\n";
        cout << "4. Exit\n";
    }
}

```

```
cout << "Enter your choice: ";
cin >> choice;

switch (choice)
{
case 1:
    insert();
    break;

case 2:
    deleteElement();
    break;

case 3:
    display();
    break;

case 4:
    cout << "Program ended\n";
    break;

default:
    cout << "Wrong choice\n";
}

} while (choice != 4);

return 0;
}
```

Terminal

Code + ▾

```
PS C:\Users\jainm\OneDrive\Desktop\dsa lab> cd "c:\Users\jainm\OneDrive\Desktop\dsa lab\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
```

QUEUE MENU

1. Insert
2. Delete
3. Display
4. Exit

Enter your choice: 1

Enter value: 10

Element inserted

QUEUE MENU

1. Insert
2. Delete
3. Display
4. Exit

Enter your choice: 1

Enter value: 20

Element inserted

QUEUE MENU

1. Insert
2. Delete
3. Display
4. Exit

Enter your choice: 1

Enter value: 30

Element inserted

Terminal

Code + ▾ 🗑️ ...

PS C:\Users\jainm\OneDrive\Desktop\dsa lab> cd "c:\Users\jainm\OneDrive\Des

Enter your choice: 1

Enter value: 30

Element inserted

QUEUE MENU

1. Insert

2. Delete

3. Display

4. Exit

Enter your choice: 3

Queue: 10 20 30

QUEUE MENU

1. Insert

2. Delete

3. Display

4. Exit

Enter your choice: 2

Deleted element: 10

QUEUE MENU

1. Insert

2. Delete

3. Display

4. Exit

Enter your choice: 3

Queue: 20 30

QUEUE MENU

1. Insert

2. Delete

3. Display

4. Exit

Enter your choice: 4

Program ended

PS C:\Users\jainm\OneDrive\Desktop\dsa lab>

Q2. Write a menu driven program with 4 options (Insert, Delete, Display, and Exit) to demonstrate the working of Circular Queues (arrays.)

```
#include <iostream>
using namespace std;

#define SIZE 5

int queue[SIZE];
int front = -1;
int rear = -1;

void insert()
{
    int x;

    if ((rear + 1) % SIZE == front)
    {
        cout << "Queue is full\n";
        return;
    }

    cout << "Enter value: ";
    cin >> x;

    if (front == -1)
    {
        front = 0;
        rear = 0;
    }
    else
    {
        rear = (rear + 1) % SIZE;
    }

    queue[rear] = x;

    cout << "Element inserted\n";
}

void deleteElement()
{
    if (front == -1)
    {

```

```

        cout << "Queue is empty\n";
        return;
    }

    cout << "Deleted element: " << queue[front] << endl;

    if (front == rear)
    {
        front = -1;
        rear = -1;
    }
    else
    {
        front = (front + 1) % SIZE;
    }
}

void display()
{
    if (front == -1)
    {
        cout << "Queue is empty\n";
        return;
    }

    cout << "Circular Queue: ";

    int i = front;

    while (true)
    {
        cout << queue[i] << " ";

        if (i == rear)
        {
            break;
        }

        i = (i + 1) % SIZE;
    }

    cout << endl;
}

```

```
int main()
{
    int choice;

    do
    {
        cout << "\nMENU \n";
        cout << "1. Insert\n";
        cout << "2. Delete\n";
        cout << "3. Display\n";
        cout << "4. Exit\n";

        cout << "Enter your choice: ";
        cin >> choice;

        switch (choice)
        {
            case 1:
                insert();
                break;

            case 2:
                deleteElement();
                break;

            case 3:
                display();
                break;

            case 4:
                cout << "Program ended\n";
                break;

            default:
                cout << "Wrong choice\n";
        }

    } while (choice != 4);

    return 0;
}
```

Terminal

Code + - [] [] ...

```
PS C:\Users\jainm\OneDrive\Desktop\dsa lab> cd "c:\Users\jainm\OneDrive\Desktop\dsa lab\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
```

MENU

1. Insert
2. Delete
3. Display
4. Exit

Enter your choice: 1

Enter value: 10

Element inserted

MENU

1. Insert
2. Delete
3. Display
4. Exit

Enter your choice: 1

Enter value: 20

Element inserted

MENU

1. Insert
2. Delete
3. Display
4. Exit

Enter your choice: 1

Enter value: 30

Element inserted

Terminal

Code + - [] [] ...

```
PS C:\Users\jainm\OneDrive\Desktop\dsa lab> cd "c:\Users\jainm\OneDrive\Desktop\d
```

MENU

1. Insert
2. Delete
3. Display
4. Exit

Enter your choice: 3

Circular Queue: 10 20 30

MENU

1. Insert
2. Delete
3. Display
4. Exit

Enter your choice: 2

Deleted element: 10

MENU

1. Insert
2. Delete
3. Display
4. Exit

Enter your choice: 3

Circular Queue: 20 30

MENU

1. Insert
2. Delete
3. Display
4. Exit

Enter your choice: 1

Enter value: 40

Element inserted

Terminal

Code + - [] [] ...

```
PS C:\Users\jainm\OneDrive\Desktop\dsa lab> cd "c:\Users\jainm\OneDrive\Desktop\dsa lab"
Circular Queue: 20 30
```

MENU

1. Insert
2. Delete
3. Display
4. Exit

Enter your choice: 1

Enter value: 40

Element inserted

MENU

1. Insert
2. Delete
3. Display
4. Exit

Enter your choice: 1

Enter value: 50

Element inserted

MENU

1. Insert
2. Delete
3. Display
4. Exit

Enter your choice: 3

Circular Queue: 20 30 40 50

MENU

1. Insert
2. Delete
3. Display
4. Exit

Enter your choice: 4

Program ended

PS C:\Users\jainm\OneDrive\Desktop\dsa lab>